

Comprehensive Architecture and Simulation of Modern Radar Systems

Introduction to Radar Systems Engineering

The foundational premise of Radio Detection and Ranging (RADAR) has remained remarkably consistent since its early implementations in the mid-twentieth century: the transmission of electromagnetic energy and the precise measurement of its reflection to determine the state of a remote object. However, the modern radar ecosystem is an intricately woven tapestry of quantum-level semiconductor physics, advanced digital signal processing, statistical probability algorithms, and high-fidelity simulated environments. Modern radar arrays are routinely required to detect exceedingly small targets against massive backgrounds of sea or land clutter, demanding sophisticated hardware and deterministic software architectures.¹

At its core, a radar system extracts a specific set of observables from the physical environment. These primary observables include target range (derived from the time delay of the returned signal), target angles in azimuth and elevation (derived from the antenna's directional orientation and phase differentials), target size (measured as Radar Cross Section, or RCS), target speed (calculated via the Doppler frequency shift), and target features for high-resolution imaging applications.² The progression of these systems is extensively documented in academic curricula such as the Massachusetts Institute of Technology (MIT) Lincoln Laboratory's "Introduction to Radar Systems" lectures, which highlight the critical transition from early mechanical dipole arrays—such as the 20-30 MHz Chain Home systems of the 1940s, which utilized 10-15 meter wavelengths and 350 kW peak power—to the agile, multi-function active phased arrays of the contemporary era.²

This report presents an exhaustive technical analysis of modern radar systems, their fundamental operational parameters, the underlying physics of lobe formation and signal propagation, and their physical deployment on airborne platforms. Furthermore, it details the digital interconnectedness of these systems via tactical datalinks such as Link 16 and satellite relays. Finally, the report investigates how these complex physical realities are modeled in deterministic software environments, specifically focusing on the Mixed Reality Simulation Platform (MIXR, formerly OpenEagles), examining the underlying C++ class architectures required for accurate real-time and constructive simulation.⁵

Electromagnetic Theory: Radio Frequency, Gain, and Antenna Lobe Formation

The efficacy of any radar system is fundamentally bottlenecked by its antenna, which serves as the physical interface between the system's internal electronic circuitry and the external electromagnetic environment. The radiation pattern of an antenna describes the directional

and angular dependence of the field strength, and sometimes the phase, of the radio waves emitted from the source.⁷

The Physics of Lobe Formation and Interference

In an ideal isotropic radiator, electromagnetic energy propagates equally in all spherical directions. However, functional radar requires highly directional energy to determine angular coordinates accurately and to maximize the energy concentrated on a specific target. This is achieved through the physical dimensions and geometry of the antenna. The radiation pattern typically consists of a primary "main lobe"—the region containing the maximum radiated power—and multiple "sidelobes" and "backlobes," which represent leaked energy propagating in unintended directions.⁷

The physics of lobe formation relies entirely on the principles of constructive and destructive interference. In a phased array antenna, the system consists of a grid of individual radiating elements. When the electromagnetic waves emitted from these individual elements are perfectly in phase at a specific observation point in the far-field (the Fraunhofer region), they interfere constructively, creating the main lobe.⁹ Conversely, in other directions, the phase differentials of the waves cause destructive interference, resulting in nulls where radiated power drops to near zero. Sidelobes form in regions where the interference is only partially destructive.¹¹

The mathematical derivation of this phenomenon in array antennas is governed by the Array Factor (AF). The total radiation pattern of an array is the product of the radiation pattern of a single element and the Array Factor. To electronically steer the main beam without mechanically moving the antenna, a specific time delay (or phase shift) is applied to the signal

at each element. The phase shift $\Delta\phi$ required between adjacent elements separated by a distance d to steer the beam to an angle θ relative to the broadside (the direction perpendicular to the array face) is calculated as:

$$\Delta\phi = \frac{2\pi d}{\lambda} \sin(\theta)$$

where λ is the operational wavelength. A critical constraint in phased array design is the prevention of "grating lobes"—which are highly undesirable secondary main lobes that cause severe spatial ambiguity. To scan over all space without creating grating lobes, the separation distance d between antenna elements in both dimensions must be kept strictly below $\lambda/2$.¹²

Sidelobe Management, Tapering, and Mutual Coupling

Sidelobes present severe tactical and functional vulnerabilities. For a receiving radar, high sidelobes allow heavy ground clutter or adversarial jamming signals to enter the receiver chain from off-axis angles, masking true targets.¹ For a transmitting radar, sidelobes waste energy

and increase the probability of detection by hostile electronic support measures (ESM) or anti-radiation missiles.

To mitigate sidelobes, radar engineers utilize amplitude tapering (or windowing) across the array. Instead of feeding every antenna element with equal power (a uniform or rectangular distribution), elements near the center of the array are fed more power than those at the edges. Tapering functions, such as the Hamming window, Taylor distributions, or adaptive reduced-rank constrained constant modulus algorithms, effectively suppress sidelobe levels.¹⁰ While a rectangular window yields the narrowest possible main lobe, its relative sidelobe peaks are substantial; applying a Hamming window broadens the main lobe slightly but dramatically reduces the amplitude of the sidelobes.¹⁰

Furthermore, engineers must strictly account for mutual coupling—the electromagnetic interaction between adjacent antenna elements. Because array elements are spaced closer than half a wavelength, the near-field radiation of one element directly alters the input impedance of its neighbors.¹² Mutual coupling creates a dependency between the antenna's impedance and its scan angle, greatly complicating power delivery. If poorly managed, it can result in "scan blindness," a phenomenon where no power is radiated at certain steering angles because the energy is entirely reflected back into the transmitter feed.¹²

The Radar Range Equation and Atmospheric Propagation

The theoretical maximum detection range of a radar system is described by the radar range equation, a fundamental deterministic model that ties together the system hardware, the target characteristics, and the physical medium of propagation.

The basic form of the radar equation for a single transmitted pulse is given by:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$$

Where P_r is the received power, P_t is the transmitted peak power, G is the antenna gain, λ is the wavelength, σ is the target's radar cross section (RCS), and R is the distance to the target. Because the energy expands spherically twice (once on the journey to the target, and once on the return journey to the receiver), the received power is inversely proportional to the fourth power of the range ($1/R^4$), representing a severe physical constraint.¹⁵

To calculate the maximum theoretical range (R_{max}), the equation is restructured to isolate R and substitute P_r with the minimum detectable signal (S_{min}), which is fundamentally a function of thermal noise. The equation becomes dependent on the signal-to-noise ratio (SNR)

required for detection:

$$R_{\max} = \left(\frac{1}{4} \right)$$

Here, A_e is the effective aperture area, k is Boltzmann's constant, T_0 is the standard system temperature, B is the receiver bandwidth, and F is the noise figure of the receiver.¹⁶

Environmental Propagation and Attenuation Factors

The standard radar equation assumes propagation in a vacuum, but operational radars function within the Earth's troposphere, necessitating complex mathematical corrections for atmospheric attenuation, diffraction, and refraction. The MIT Lincoln Laboratory propagation models highlight four primary environmental effects that fundamentally alter radar performance¹⁷:

1. **Refraction and the 4/3 Earth Radius Model:** The index of refraction of the atmosphere decreases with altitude, causing radar beams to bend downward slightly. To simplify calculations, engineers treat radar rays as traveling in straight lines but artificially increase the Earth's radius by a factor of 4/3. This accounts for the standard atmospheric bending and accurately places the "radar horizon" slightly further than the geometric optical horizon.¹⁷
2. **Ducting:** Under certain weather conditions, such as sharp temperature inversions, the refractive index drops so drastically that radar beams are trapped in an atmospheric waveguide, or "duct." This allows for anomalous detection at vastly extended over-the-horizon ranges, though it leaves critical coverage gaps directly above the duct layer.¹⁷
3. **Multipath Reflection:** When detecting low-altitude targets, the radar receives both the direct echo from the target and a delayed echo that bounces off the Earth's surface (ground or sea). The reflection coefficient (Γ) often approaches -1 for shallow grazing angles.¹⁷ The phase difference between the direct and reflected waves causes alternating constructive and destructive interference, resulting in a lobing effect in the vertical coverage diagram. A target caught in a null (destructive interference) will momentarily disappear from the radar display.¹⁷
4. **Diffraction:** Lower frequency radars (e.g., L-band, 1-2 GHz) diffract over the curvature of the earth and terrain obstacles far better than higher frequency radars (e.g., X-band, 8-12 GHz), offering slightly better performance against targets hiding just below the horizon line. Specific MIT Lincoln Lab models demonstrate an 80 dB loss for X-band versus a 60 dB loss for L-band due to diffraction at a 100-foot altitude over 60 kilometers.¹⁷

Hardware Engineering: AESA Manufacturing and T/R Module Integration

The evolution from mechanically steered arrays (MSA) and passive electronically scanned arrays (PESA) to the Active Electronically Scanned Array (AESA) marks the most significant architectural shift in modern radar engineering. In an AESA, the single central high-power transmitter (such as a Traveling Wave Tube or Klystron) is eliminated. Instead, each antenna element—or small group of elements—is coupled with an individual solid-state Transmit/Receive (T/R) module.¹⁹

An AESA radar for a modern fighter aircraft may contain over a thousand T/R modules, each functioning as an independent radar unit. This distributed gain architecture yields profound tactical advantages: it eliminates the catastrophic single-point-of-failure inherent to central transmitters, permits the simultaneous formation of multiple independent beams at different frequencies, and dramatically enhances resistance to jamming by spreading signal emissions across a wide, unpredictable spectrum.²⁰

T/R Module Components and Internal Architecture

T/R modules are complex, tightly integrated micro-electronic packages responsible for signal amplification and phase control. A standard T/R module is bifurcated into a transmit path and a receive path.¹⁹

Component	Functional Path	Engineering Description and Role
Power Amplifier (PA)	Transmit	Amplifies the low-power excitation signal to the required output power for radiation. PAs are often cascaded (e.g., a three-stacked structure) to achieve the necessary gain before transmission. ¹⁴
Low Noise Amplifier (LNA)	Receive	Amplifies the exceedingly weak target echo immediately upon reception, establishing the noise figure of the entire receiver chain before cable losses can degrade the signal. ¹⁴
Phase Shifter	Common	Provides the programmable

		phase delay (typically 6-bit or 8-bit digital control) required for electronic beam steering without moving parts. ¹⁹
Variable Attenuator	Common	Allows for amplitude control, enabling array tapering for sidelobe suppression and adaptive beamforming algorithms. ²³
Circulator/Duplexer	Junction	Isolates the highly sensitive receive path from the high-power transmit path while sharing a single antenna element. It prevents load modulation of the PA caused by impedance changes during beam scanning. ¹⁹

The progression from purely-analog-RF to digital intermediate frequencies (IF) represents the next generation of AESA advancement. Modern T/R modules increasingly utilize high-speed digital Serializer/Deserializer (SerDes) interfaces and integrated Analog-to-Digital Converters (ADCs) directly at the element level, reducing system complexity and facilitating advanced digital beamforming.¹⁸

Manufacturing Processes and Materials Science

The manufacturing of T/R modules constitutes up to 50% of the cost of an AESA system.¹⁹ Consequently, production base optimization is highly prioritized by prime contractors.²⁶ Historically, monolithic microwave integrated circuits (MMICs) within T/R modules were manufactured using Gallium Arsenide (GaAs). While effective, GaAs suffers from limited power density and thermal conductivity constraints.²⁷

The modern manufacturing paradigm has decisively shifted toward Gallium Nitride (GaN) grown on Silicon Carbide (SiC) substrates.²⁹ GaN offers a significantly wider electron bandgap and higher breakdown voltage compared to GaAs, enabling power densities up to ten times higher.²⁷ This allows radars to output massive peak power without expanding the physical aperture, an essential metric for Size, Weight, and Power (SWaP) constrained platforms such as Unmanned Aerial Vehicles (UAVs) and fighter aircraft.²⁴

However, the immense power density of GaN generates substantial localized heat, creating

complex thermal management challenges. Modern manufacturing employs Low-Temperature Co-fired Ceramic (LTCC) packaging, which allows for three-dimensional wiring, embedded passives, and excellent thermal matching.²⁷ To prevent mechanical stress fractures caused by fluctuating temperatures, GaN power devices are soldered onto advanced heatsink materials designed to perfectly match the Coefficient of Thermal Expansion (CTE) of the MMIC.²⁷

Vibration presents another critical failure vector for PCB enclosures within the radar housing. Recent advancements include the integration of single-particle impact dampers (SPIDs) within the PCB enclosure. By utilizing a linear contact model to mimic particle-structure interactions, engineers can drastically reduce the amplitude ratio of physical vibrations, preventing micro-fractures in the T/R module's internal wiring.³²

Signal Processing: Dwell Time, Pulse Integration, and Target Fluctuation

Because of the inverse fourth-power loss of range dictated by the radar equation, a single radar pulse is rarely sufficient to breach the detection threshold in the presence of thermal noise and background clutter. Signal processors overcome this by transmitting a sequence of pulses in a specific direction—an interval known as the "dwell time"—and integrating the returning echoes. Pulse integration improves the SNR by averaging out random noise while reinforcing the consistent target signal.¹⁶

Coherent vs. Non-Coherent Integration

The mathematical method of integration fundamentally affects the performance, sensitivity, and hardware requirements of the radar:

1. **Coherent Integration:** In this pre-detection method, the integrator acts on the intermediate frequency (IF) signals, preserving both the amplitude and the phase of the incoming pulses. Utilizing digital Fast Fourier Transforms (FFTs), the processor accumulates the signal vectors.³⁴ Coherent integration is highly efficient, providing an SNR enhancement directly proportional to N (the total number of pulses integrated).³⁴ However, if a target exhibits rapid phase fluctuations from pulse to pulse, coherent integration can destructively interfere, yielding no benefit.³⁴
2. **Non-Coherent Integration:** In this post-detection method, the integrator sits after the envelope detector. Phase information is entirely discarded, and only the amplitudes (or powers) of the pulses are summed.³⁴ The efficiency of non-coherent integration grows roughly as $\sqrt{N}$. While theoretically less optimal than coherent integration for highly stable targets, it remains robust and effective for rapidly fluctuating targets.³⁴

Binary Pulse Integration (BPI) Algorithm

To balance computational processing overhead and detection probability, many modern radar

systems utilize Binary Pulse Integration (BPI), also known as M -of- N detection or dual-threshold detection.³⁴ Rather than summing raw voltages, BPI utilizes discrete logic circuits.³⁴

In a standard BPI scheme, the radar sets a primary amplitude threshold for individual pulses. If a single pulse exceeds this threshold, it is logged in the processing buffer as a binary "1"; if not, it is logged as a "0".³⁶ The system then evaluates N pulses received over the entirety of the dwell time. If at least M pulses out of N cross the primary threshold, a secondary threshold is met, and a firm target detection is declared.³⁴

The probability of declaring a detection (P_D) using BPI is determined mathematically by the binomial distribution:

$$P_D = \sum_{k=M}^N \binom{N}{k} p^k (1-p)^{N-k}$$

where p is the probability of a single pulse crossing the primary threshold. BPI is highly resilient to transient clutter spikes and active jamming pulses, as an anomalous interference spike will only register as a single "1", failing to meet the rigorous M -of- N criteria for target declaration. The optimal ratio of M to N depends heavily on the target's fluctuation characteristics and the desired probability of false alarm.³⁵

Radar Cross Section (RCS) and Target Fluctuation Models

The Radar Cross Section (σ) is the hypothetical area required to intercept the incident power density and isotropically scatter it to produce the echo power observed at the receiver. RCS is not simply equivalent to a target's physical size; it is a complex electromagnetic property dictated by the object's precise geometry, material composition (e.g., radar-absorbing materials), and the incident angle of the radar beam.³⁸ Targets explicitly designed for stealth rely on faceted geometries that scatter energy away from the monostatic receiver or employ specialized materials that absorb microwave energy, converting it into heat.³⁸

Because real targets are complex geometries consisting of multiple scatterers (e.g., engine turbines, wing leading edges, tail fins), tiny changes in the target's aspect angle relative to the radar result in massive phase shifts between the individual scattering centers. This causes the total RCS to scintillate rapidly from pulse to pulse or scan to scan. To mathematically model this behavior for detection probability calculations, engineers utilize the Swerling target models³⁴:

Swerling Model	Fluctuation Rate	Target Description	Probability Density Function (PDF)
Case 0 (or 5)	None	Ideal, stable target (e.g., a calibration sphere).	Delta function
Case 1	Scan-to-Scan	Many independent scatterers of similar size (e.g., an aircraft in highly stable flight).	Rayleigh (Exponential for RCS)
Case 2	Pulse-to-Pulse	Many independent scatterers, rapid motion (e.g., a rapidly maneuvering aircraft).	Rayleigh
Case 3	Scan-to-Scan	One dominant scatterer plus many small scatterers.	Chi-squared (4 degrees of freedom)
Case 4	Pulse-to-Pulse	One dominant scatterer plus many small scatterers, rapid motion.	Chi-squared (4 degrees of freedom)

Airborne Targeting, Clutter Mitigation, and Space-Based Radar Limitations

The deployment medium of a radar array fundamentally alters its design parameters and signal processing requirements. Ground-based radars face different challenges than airborne or

space-based platforms.

Airborne Radar Dynamics and Clutter Mitigation

Airborne platforms, such as the U.S. Air Force E-3 Sentry AWACS or modern fighter aircraft (e.g., F-22, F-35), operate within the troposphere, granting them a localized radar horizon of several hundred miles.²¹ Fighter aircraft utilize multi-function AESA radars in their nose cones to seamlessly alternate between air-to-air targeting, synthetic aperture ground mapping, and even focused electronic attack (jamming) using the high-power density of their transmit modules.²⁰ For aircraft targeting and tracking, the radar must dynamically manage its beam schedule, balancing rapid, wide-volume search scans with focused dwell times on established tracks, all while the host aircraft undergoes extreme kinetic maneuvers.²¹

A critical challenge for airborne radar is mitigating ground clutter. Because the radar is looking down, the target echo is often buried in massive radar returns from the earth's surface. As detailed in the MIT Lincoln Laboratory coursework, engineers overcome this using Moving Target Indicator (MTI) and Pulse Doppler techniques.¹ Pulse Doppler processing utilizes highly sensitive coherent integration to isolate the Doppler frequency shift of a moving aircraft from the stationary frequency of the ground clutter.¹ However, this processing must account for the host aircraft's own velocity vector, which induces a Doppler shift on the entire clutter field, requiring complex digital compensation algorithms.

The Physics Constraints of Space-Based Radar

A persistent engineering debate involves the utility and difficulty of Airborne Early Warning and Control (AWACS) systems versus Low Earth Orbit (LEO) space-based radar.¹⁵ While a satellite-based radar inherently solves the "radar horizon" problem by observing from space, it faces severe physical and geometric penalties:

1. **The Range Penalty:** As established by the radar equation, signal strength decays by $1/R^4$. Elevating a radar from a 30,000-foot aircraft to a 400-kilometer LEO satellite increases the range variable by an order of magnitude, drastically crushing the received signal strength.¹⁵
2. **Power Generation and Thermal Shedding:** To overcome the massive $1/R^4$ loss, a satellite radar must output enormous power. Generating this power via solar panels and—more critically—shedding the resultant waste heat in the vacuum of space (where convection is impossible, leaving only radiative cooling) requires massive, heavy radiators that are difficult to launch.¹⁵
3. **Target Geometry:** Terrestrial or airborne radars typically view aircraft from horizontal or low-elevation angles, where structures like vertical stabilizers and engine nacelles act as highly reflective corner reflectors. A space-based radar looks straight down. The top of an aircraft wing is a flat, poorly reflective surface, further minimizing the effective RCS and exacerbating the range loss.⁴¹

Consequently, while space-based surveillance is highly effective for slow-moving ground target indication or large maritime tracking using Synthetic Aperture Radar (SAR), real-time tactical targeting of highly maneuverable, low-RCS fighter platforms relies predominantly on airborne nodes.

Network-Centric Warfare: Datalinks and Radar Interoperability

In modern battlespaces, the radar picture generated by a single platform is rarely utilized in isolation. The integration of distributed sensors into a unified Common Operating Picture (COP) is the foundational doctrine of network-centric warfare. The primary conduits for this interoperability are Tactical Data Links (TDL), most notably Link 16 and the Multifunction Advanced Data Link (MADL).⁴²

Link 16 Architecture and Time Division Multiple Access

Link 16 is a highly structured, jam-resistant, encrypted digital network standardized under MIL-STD-6016 and NATO STANAG 5516.⁴⁶ Unlike traditional point-to-point voice radios, Link 16 operates on a Time Division Multiple Access (TDMA) architecture. The L-band frequency spectrum (969 – 1206 MHz) is divided into rapidly shifting time slots. Network participants are assigned specific time slots (up to 128 slots per second) to broadcast or receive data omnidirectionally.⁴²

A unique characteristic of Link 16 is its nodeless architecture. The network does not rely on a central server or command router; all Link 16-capable terminals (such as the MIDS-LVT terminal) act as independent nodes, maintaining network integrity even if primary command elements are destroyed.⁴⁴

J-Series Messages and Network Participation Groups

Data injected into the Link 16 network—such as radar tracks generated by an AWACS—is formatted into strictly defined binary "J-series" messages.⁴⁵ These messages are categorized into Network Participation Groups (NPGs) based on specific functional areas:

NPG Designation	Primary Function	Relevant J-Series Messages	Description
NPG 5 & 6	PPLI	J2.2 (Air), J2.5 (Land Point)	Precise Participant Location and Identification. Friendly platforms broadcast their

			own GPS location, altitude, and system status. ⁴⁶
NPG 7	Surveillance	J3.2 (Air Track), J3.5 (Land Track)	Raw or correlated radar tracks observed by sensors, shared seamlessly to populate the COP across the theater. ⁵⁰
NPG 8 & 9	Command / Control	J12 (Control)	Target assignments, vectoring, and weapon coordination from C2 nodes to individual fighters. ⁴⁶
NPG 10	Electronic Warfare	J3.7 (EW Product)	Sharing critical data on adversarial radar emissions and jamming environments. ⁴⁶

This architecture enables revolutionary tactics. If an AWACS establishes a firm radar track on an adversarial asset, it generates a J3.2 Air Track message and broadcasts it on NPG 7. A stealth fighter operating with its own AESA radar deactivated to maintain emissions control (EMCON) receives this J3.2 message, visualizing the target on its displays solely through the Link 16 integration.³⁹ Furthermore, missile datalink updates are often transmitted using the mechanical or AESA radar's side lobes, allowing the host aircraft to guide a lofted missile without directly pointing the main beam at it.⁵²

Beyond Line of Sight (BLOS) and Satellite Relays

A fundamental physical limitation of Link 16 is its reliance on L-band radio waves, which are strictly limited to Line of Sight (LOS)—typically 300 nautical miles.⁴⁷ Traditionally, Joint Range Extension Applications Protocol (JREAP-A) was used to encapsulate J-series messages over separate serial satellite links to bridge continents, requiring secondary transmission hardware.⁴⁵

However, groundbreaking developments by the Space Development Agency (SDA) have successfully demonstrated direct Link 16 communication via space. Utilizing the Proliferated Warfighter Space Architecture (PWSA) satellite constellation in Low Earth Orbit (specifically

York Space Systems' T0 Transport Layer satellites), test units have transmitted raw Link 16 J-series messages directly from orbiting satellites down to terrestrial receivers and airborne fighters.⁵³ This technological leap completely bypasses the terrestrial LOS limitation without requiring secondary conversion protocols, meshing seamlessly with highly directional datalinks like MADL to ensure persistent data fusion across conventional and stealth platforms globally.⁵³

Software Implementation: Simulating Radar in the MIXR Framework

To engineer, test, and refine both the physical radar architectures and their complex datalink integrations without the prohibitive cost of live-fly testing, the defense industry relies heavily on robust, high-fidelity simulation frameworks. One prominent environment is the Mixed Reality Simulation Platform (MIXR, previously known as OpenEagles/EAAGLES), a mature, open-source C++ framework specifically designed to support the rapid construction of distributed, deterministic real-time simulations.⁵

MIXR heavily utilizes the Model-View-Controller (MVC) and Component design patterns. It is deeply integrated with military networking standards such as Distributed Interactive Simulation (DIS) and High-Level Architecture (HLA), allowing simulated entities to interact seamlessly across wide-area networks.⁶

The Core C++ Class Hierarchy

The architecture of a simulated radar in MIXR maps directly to an object-oriented class hierarchy that models the physical flow of radio frequency energy and data processing⁵⁷:

1. **LcObject**: The foundational root class for all objects in the framework. It handles dynamic memory management, reference counting (garbage collection), and provides runtime type information essential for the framework's factory system.⁵⁷
2. **LcComponent**: Inherits from LcObject. This is the fundamental building block for all hierarchical simulation models. It manages lists of subcomponents, facilitating a "Push" architecture where data is actively sent down the tree (e.g., from an aircraft to its radar) rather than subcomponents having to query or pull it.⁵⁷
3. **RfSystem**: Represents any generic radio frequency system, managing baseline RF physics variables like frequency, transmission power, and wave propagation algorithms.⁵
4. **Sensor**: Inherits from LcComponent. Contains logic generic to all optical, IR, and RF sensors, such as establishing lines of sight and maintaining tracking lists of detected entities.⁵⁷
5. **Radar**: The specific instantiation of an active emitting sensor. It inherits from Sensor and implements the radar range equation.⁵⁷
6. **Antenna**: Handled as a distinct subcomponent of the radar, simulating beam steering coordinates, antenna gain profiles, and main lobe/sidelobe shaping.⁵⁶

Framework users build upon these core classes using standardized C++ macros. For instance, creating a new, specialized radar model begins in the header file with the `DECLARE_SUBCLASS(Radar, Sensor)` macro, which establishes the necessary boilerplate inheritance and type definitions, and is implemented via the `IMPLEMENT_SUBCLASS(Radar, "Radar")` macro in the source file, which maps the class to the framework's configuration parser.⁵⁷

Deterministic Real-Time Execution: `updateTC` and `updateData`

For human-in-the-loop (Virtual) and hardware-in-the-loop simulations, the radar model must execute with strict determinism; frame-rate drops or latency spikes invalidate the simulation physics and corrupt the data analysis. To achieve this, MIXR utilizes a highly optimized cyclic scheduler that partitions processing into two distinctly separated threading execution models: the foreground thread and the background thread.⁶

All simulation classes inheriting from `LcComponent` must implement two primary polymorphic update methods⁵⁷:

1. **`updateTC(LCreal deltaTime)`**: This is the "Time Critical" method. It executes in the high-priority foreground real-time thread. In the Radar class, `updateTC` contains the strict, deadline-driven mathematical modeling of the radar. This includes calculating the radar range equation, checking target boundary limits, computing phase shifts for the Antenna subcomponent's beam steering, and executing the Binary Pulse Integration logic matrix. A strict rule within the MIXR framework is that `updateTC` code must *never* block or wait for external I/O (like network calls or file reads), as doing so would cause the real-time frame to overrun its allocated delta-time threshold.⁶
2. **`updateData(LCreal deltaTime)`**: This executes in the lower-priority background thread. It is designed for operational tasks without strict, microsecond timing constraints. For a radar component, `updateData` manages the formatting of internal track data for external display screens (like a multi-function display in a simulated cockpit), processing data logs, and initiating the sending of DIS/HLA network packets (such as translating an internal radar track into a simulated Link 16 J3.2 message).⁵⁷

Simulating the Dwell, BPI, and RCS Interfacing

When the cyclic scheduler triggers Phase 2 (the specific phase reserved for RF sensor calculations), the Radar class initiates its complex logic.⁶ The radar requests its Antenna subcomponent to position the simulated beam based on the current search pattern or track schedule. The system then iterates through the list of potential targets provided by the simulation environment.

A critical step is determining the target's aspect angle relative to the radar beam. To accurately model target dynamics, MIXR frequently interfaces with external flight dynamics models like JSBSim, an open-source flight dynamics engine.⁵⁵ JSBSim provides continuous

6-degree-of-freedom orientation data. The radar utilizes this orientation to reference the target's configured RCS profile. Depending on the fidelity required by the simulation, this might be a simple constant cross-section, or it may utilize algorithms like the Correlated Statistical (CORS) model or interpolation tables to induce complex statistical noise based on the Swerling models.⁶⁰

The radar then calculates the signal-to-noise ratio using the deterministic radar equation within updateTC. If the target's RCS is fluctuating, the returned power is modified dynamically. The radar object subsequently executes its configured pulse integration routine, utilizing variables like the bpi parameter to track M -of- N threshold logic over the simulated dwell time.⁶¹ If the primary threshold is crossed sufficiently during the simulated dwell time, a firm track is instantiated in the Sensor component's internal track list. Finally, a "Push" event utilizes framework message tokens (e.g., via the event(int token) method) to notify the host platform's mission computer, effectively simulating the entire hardware and signal processing chain entirely in software.⁵⁷

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